

Creozen Ltd.

FREE WORKSHOP

RAG Pipelines (Ingestion and Inference)

A hands-on, depth-first workshop covering everything you need to build production-ready RAG systems. Learn ingestion, retrieval, ranking, evaluation, and build a complete pipeline from scratch.

SESSIONS

What is RAG? / RAG Architecture

Chunking, Embeddings & Vector Search

Retrieval, Ranking Evaluation

Hands-On: Build a RAG Pipeline

13

JULY

2 0 2 6

07.00 PM -
09.00 PM
(IST Time)



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www.creozen.co.uk

SESSION 1: WHAT IS RAG? (15 MINUTES)

TOPICS COVERED

- WHAT IS RAG?
- WHY RAG EXISTS - THE PROBLEM WITH STANDALONE LLMS
- BRIDGING STATIC MODEL KNOWLEDGE AND DYNAMIC DATA
- RAG VS FINE-TUNING - TRADE-OFFS, COST & USE CASES
- WHEN TO USE RAG, FINE-TUNE, OR COMBINE BOTH

LEARNING OUTCOME

PARTICIPANTS WILL UNDERSTAND WHY RAG EXISTS AND HOW IT COMPARES TO FINE-TUNING FOR REAL-WORLD LLM APPLICATIONS.

SESSION 2: RAG ARCHITECTURE & CHUNKING (30 MINUTES)

TOPICS COVERED

- END-TO-END RAG PIPELINE ARCHITECTURE
- DOCUMENT INGESTION TO RESPONSE GENERATION
- CHUNKING STRATEGIES: FIXED-SIZE, RECURSIVE, SEMANTIC, SLIDING WINDOW
- HOW OVERLAP AFFECTS RETRIEVAL QUALITY
- EMBEDDINGS & VECTOR SEARCH
- HOW TEXT IS CONVERTED INTO HIGH-DIMENSIONAL VECTORS
- SEMANTIC SIMILARITY SEARCH BEYOND KEYWORD MATCHING

Learning Outcome

- Participants will understand RAG architecture, chunking strategies, and how embeddings enable intelligent retrieval.

Vector Databases

- Hands-on familiarity with FAISS, ChromaDB, Pinecone, and Weaviate. Understand indexing, storage, and querying at scale.

SESSION 3: RETRIEVAL, RANKING & EVALUATION (30 MINUTES)

TOPICS COVERED

- RETRIEVAL: BM25, SEMANTIC & HYBRID APPROACHES
- RANKING & RE-RANKING WITH CROSS-ENCODERS
- RAGAS FRAMEWORK: FAITHFULNESS, ANSWER RELEVANCY, CONTEXT PRECISION, CONTEXT RECALL
- REAL-WORLD USE CASES: ENTERPRISE Q&A, LEGAL RESEARCH, HEALTHCARE, CODE DOCS

HANDS-ON SESSIONS

- BUILD A COMPLETE RAG PIPELINE FROM SCRATCH - LOAD, CHUNK, EMBED, STORE, RETRIEVE & GENERATE
- QUERY YOUR OWN DOCUMENTS - BRING YOUR OWN PDF AND BUILD A WORKING Q&A SYSTEM

SESSION 4: Q&A AND CAREER GUIDANCE (20 MINUTES)

DISCUSSION TOPICS

- RAG IN PRODUCTION - CHALLENGES & SOLUTIONS
- SCALING RAG SYSTEMS
- AI ENGINEERING CAREER OPPORTUNITIES
- BUILDING RAG PROJECTS FOR PORTFOLIO
- INDUSTRY BEST PRACTICES
- OPEN Q&A SESSION

WHO SHOULD ATTEND?

- SOFTWARE DEVELOPERS
- ML/AI ENGINEERS
- DATA SCIENTISTS
- FULL-STACK ENGINEERS
- AI ENTHUSIASTS
- ANYONE INTERESTED IN LLMS AND RAG SYSTEMS

WHAT YOU WILL LEARN

- UNDERSTAND RAG ARCHITECTURE END-TO-END
- IMPLEMENT CHUNKING, EMBEDDINGS & VECTOR SEARCH
- BUILD RETRIEVAL PIPELINES WITH BM25, SEMANTIC & HYBRID APPROACHES
- EVALUATE RAG SYSTEMS USING RAGAS FRAMEWORK
- DEPLOY PRODUCTION-READY RAG PIPELINES
- QUERY YOUR OWN DOCUMENTS WITH A CUSTOM RAG SYSTEM

Prerequisites

Basic Python programming knowledge
Familiarity with APIs and JSON
Laptop with internet connection
Curiosity to learn RAG and modern AI systems